

Scarecrow System – Design & Implementation

1. What This Project Does

We are building a high-tech scarecrow with a double-barrelled approach to keep pests out of farms.

It has two separate systems that work at the same time when an animal comes near:

1. **The Main Robot Body:** It has an electronic "eye" (PIR/IR sensor) near the ground. When it sees an animal move, the "brain" (ESP32 board) tells the arms to start waving and the speaker to make a loud noise.
 2. **The Solar Repellent:** On the very top is a separate, solar-powered gadget. It sits quietly in standby mode, using no power. It has **its own motion sensor**. When it detects an animal, it wakes up and emits **hidden ultrasonic sounds** (sounds humans can't hear but animals hate) and **flashes strong lights** to confuse and shoo them away.
-

2. Step-by-Step Build Instructions

Step 1: Build the PVC Frame

1. **Cut the Pipes:** Cut the long 1-inch pipe into these pieces: four short legs (0.8 ft each), one long body piece (1.5 ft), one neck piece (0.8 ft), and two very short shoulder pieces (0.25 ft each).
2. **Make the Base:** Take the four short legs and plug them into the **center cross connector** to make an "X" shape on the ground. Put caps on the ends of the legs.
3. **Build the Body:** Stick the long body pipe straight up into the center of the base. Put the **T-shaped connector** on top of that pipe.
4. **Add Arms & Neck:** Stick the two very short shoulder pipes into the sides of the "T". Put **elbow (corner) connectors** on the ends of those arms (this is where motors go later). Stick the neck pipe into the very top of the "T".
5. **Dress It Up:** Put the shirt on the body frame. Cut a hole in the bottom of the pot and slide it over the neck to make the head. Firmly attach the **Solar Repeller** onto the very top pipe. Stick the yellow sticky trap onto the shirt area.

Step 2: Connect the Electronics (Wiring)

1. **Power Connection:** Take the main 12V power adapter. Connect its wires to the input of the **Motor Driver Board** and the input of the small **Buck Converter** (the one that steps 12V down to 5V).
2. **Powering the Brains:** Take the 5V output from the Buck Converter and use it to power the main **MCU (ESP32)**, the **Speaker Module**, and the **Motion Sensor**.
3. **Connecting Motors:** Connect the two motor wires to the outputs on the **Motor Driver Board**. Connect the control pins from that board to the main **MCU**.
4. **Connecting Sensor & Sound:** Connect the data pin from the **Motion Sensor** to the MCU so it can send signals. Connect the audio input pin on the **Speaker Module** to the Brain Board so it can play sound.

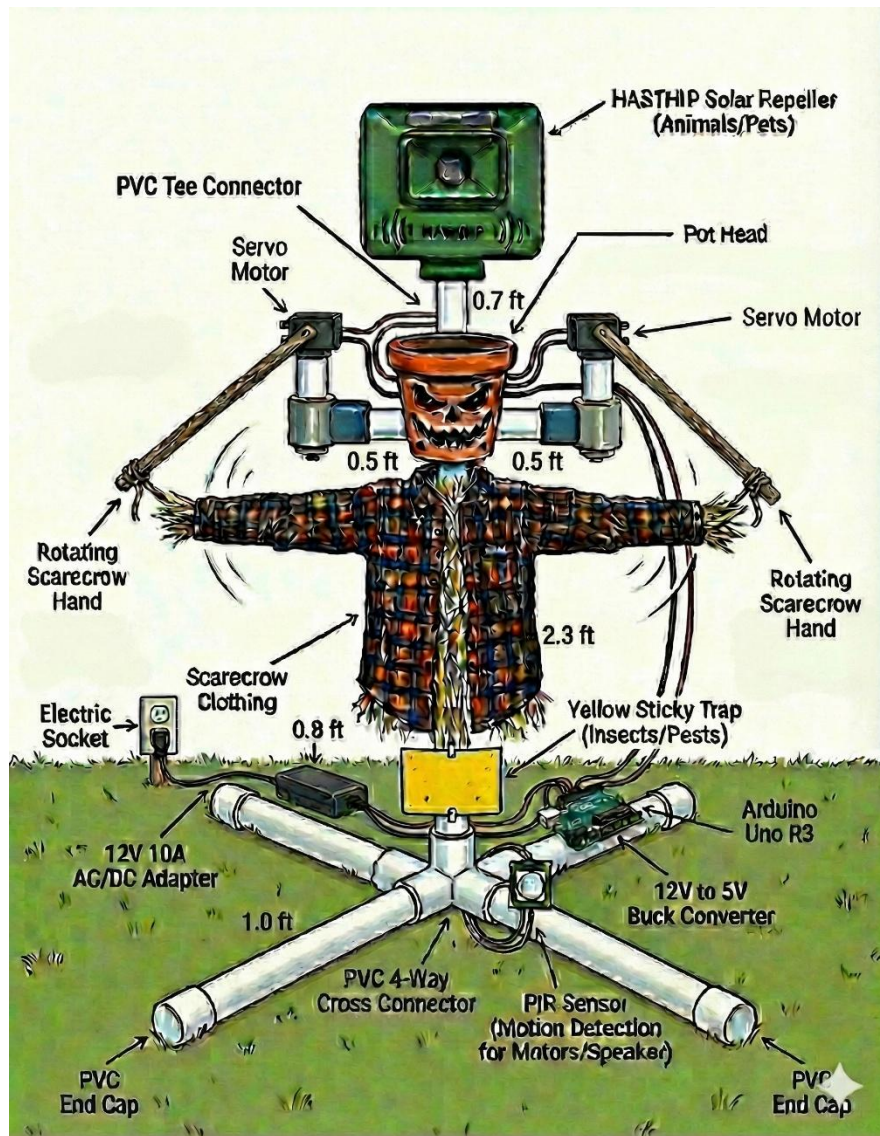
Step 3: Final Assembly

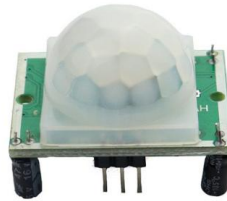
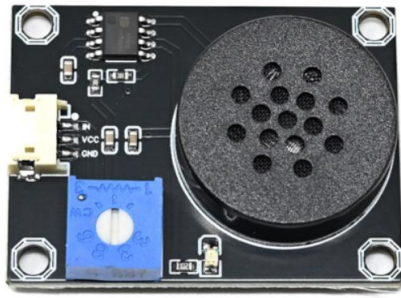
1. **Attach Motors:** Use zip-ties or screws to fasten the motors securely to the PVC elbows on the shoulders. Attach the wooden sticks to the motors so they spin like arms.
2. **Place Sensor & Speaker:** Mount the speaker on the base facing outward so it's loud. Place the motion sensor on the base pointing toward where animals might come from.

Step 4: How It Will Work (The Logic)

Once programmed, here is how the whole system reacts automatically:

1. **Wait and Watch:** Both the main body and the solar topper sit quietly, monitoring the area with their separate sensors.
2. **Detect:** When an animal approaches, both sensors will likely detect the movement at roughly the same time.
3. **React (Simultaneously):**
 - The **Main Body** turns on the motors to wave the sticks and plays a loud alarm sound for 10 seconds.
 - The **Solar Topper** wakes up and starts flashing its lights and blasting ultrasonic sounds.
4. **Reset:** After doing their job, both systems stop and go back to waiting for the next movement.





PVC Pipe Calculation (3 ft Target Height)

- **Base legs:** $4 \times 1.0 \text{ ft} = 4.0 \text{ ft}$
- **Torso pipe:** $1 \times 2.3 \text{ ft} = 2.3 \text{ ft}$
- **Shoulder arms:** $2 \times 0.5 \text{ ft} = 1.0 \text{ ft}$
- **Neck pipe:** $1 \times 0.7 \text{ ft} = 0.7 \text{ ft}$

New Total PVC Pipe Length = 8.0 feet (*buy roughly 10 feet of pipe to account for cutting waste*)

PVC Connector Calculation

- **End Caps:** 4 (For the base legs)
- **4-Way Cross:** 1 (For the base center - ensure one outlet points UP)
- **Tee Connector:** 1 (For the shoulder/neck junction)
- **90 Degree Elbows:** 2 (For connecting motor to the pipes)

Total PVC Connectors = 8 pieces

Approximate Cost Estimation / Bill of Materials (BOM)

S.No.	Component	Quantity / Details	Unit Price (Rs.)	Total Cost (Rs.)
Structure & Decor				
1.	PVC Pipe	10.0 feet total	60 / feet	600
2.	PVC Connectors	8 pieces total	30 / piece	240
3.	Plastic Pot (Head)	1 unit	150 / unit	150
4.	Yellow Sticky Trap	1 pack/unit	250 / unit	250
Electronics (Main System)				
5.	Arduino Uno R3 with Cable (Microcontroller)	1 unit	600 / unit	600
6.	12V Servo Motors (for hands)	2 units	900 / unit	1800
7.	SmartElex Digital Speaker Module	1 unit	300 / unit	300
8.	PIR Motion Sensor Module	1 unit	100 / unit	100
Power & Independent Units				
9.	HASTHIP Solar Repeller	1 unit	1300 / unit	1300
10.	12V 10A AC/DC Adapter	1 unit	500 / unit	500
11.	12V to 5V Buck Converter	1 unit	100 / unit	100
Misc.				

12.	Glue, Sticks, Marker, Jumper Wires, Scarecrow Decoration etc.	-	200	200
	GRAND TOTAL			Rs. 6,140

Note: Some components may be present in the lab